AIM:

Mote-to-Mote Radio Communication Simulation

Simulate packet transmission between two motes and analyze signal strength and packet loss under varying radio gain values.

Code:

Mote.ned:

```
package motecommunication;

simple Mote
{
    parameters:
        double sendInterval @unit(s);
    gates:
        input in;
        output out;
}
```

RadioNetwork.ned:

```
package motecommunication;

network RadioNetwork
{
    submodules:
        mote1: Mote
        {
            parameters:
                sendInterval = 1s;
        }

        mote2: Mote
        {
            parameters:
                sendInterval = 0s;
        }

    connections:
        mote1.out --> mote2.in;
        mote2.out --> mote1.in;
}
```

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Mote.cc:

```
#include <omnetpp.h>

class Mote : public cSimpleModule
{
  private:
    cMessage *timer;
    int received;

  protected:

    virtual void initialize()
    {
        received = 0;

        if(strcmp(getName(),"mote1")==0)
        {
            timer = new cMessage("sendTimer");
            scheduleAt(simTime()+1,timer);
        }
        else
            timer = NULL;
    }

    virtual void handleMessage(cMessage *msg)
    {
        if(msg==timer)
        {
            EV << "Sending Packet\n";

            cMessage *pkt=new cMessage("DataPacket");

            send(pkt,"out");

            scheduleAt(simTime()+1,timer);
        }
        else
        {
            received++;

            EV << "Packet Received Count="
               << received << "\n";

            delete msg;
        }
    }

    virtual ~Mote()
    {
        cancelAndDelete(timer);
```

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```
    }
};

Define_Module(Mote);
```

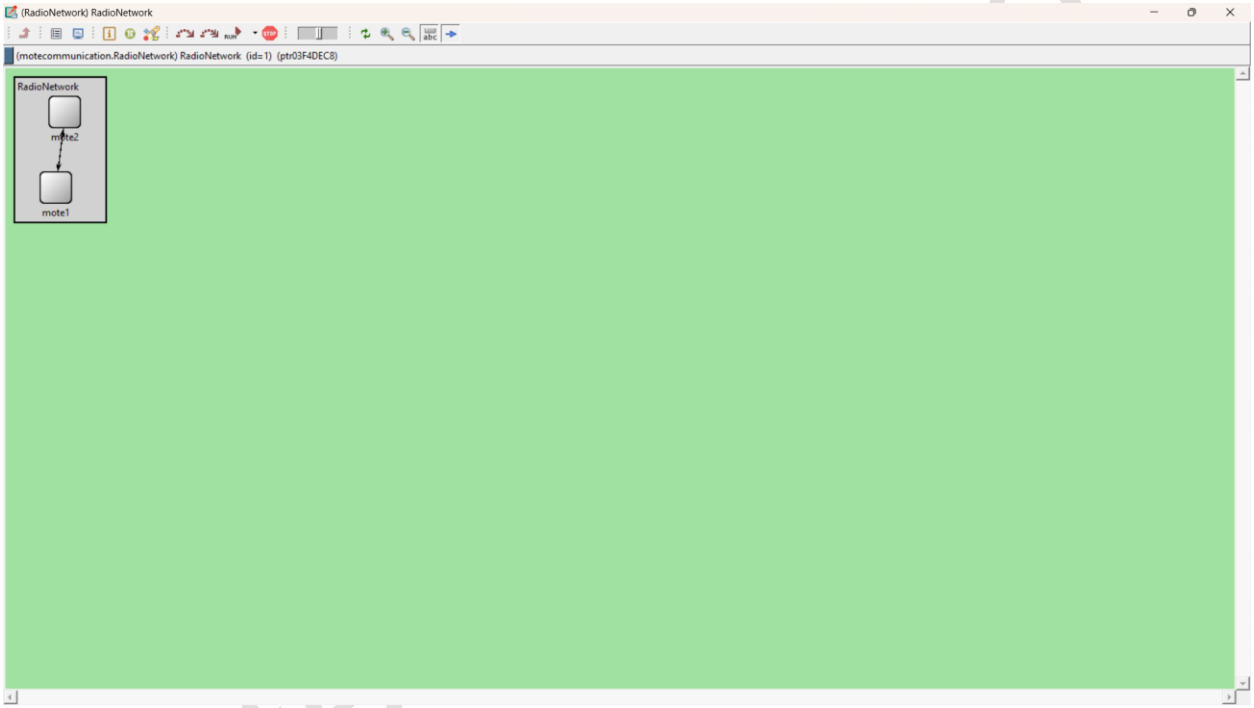
omnetpp.ini:

```
[General]
network = motecommunication.RadioNetwork

sim-time-limit = 20s
```

OUTPUT:

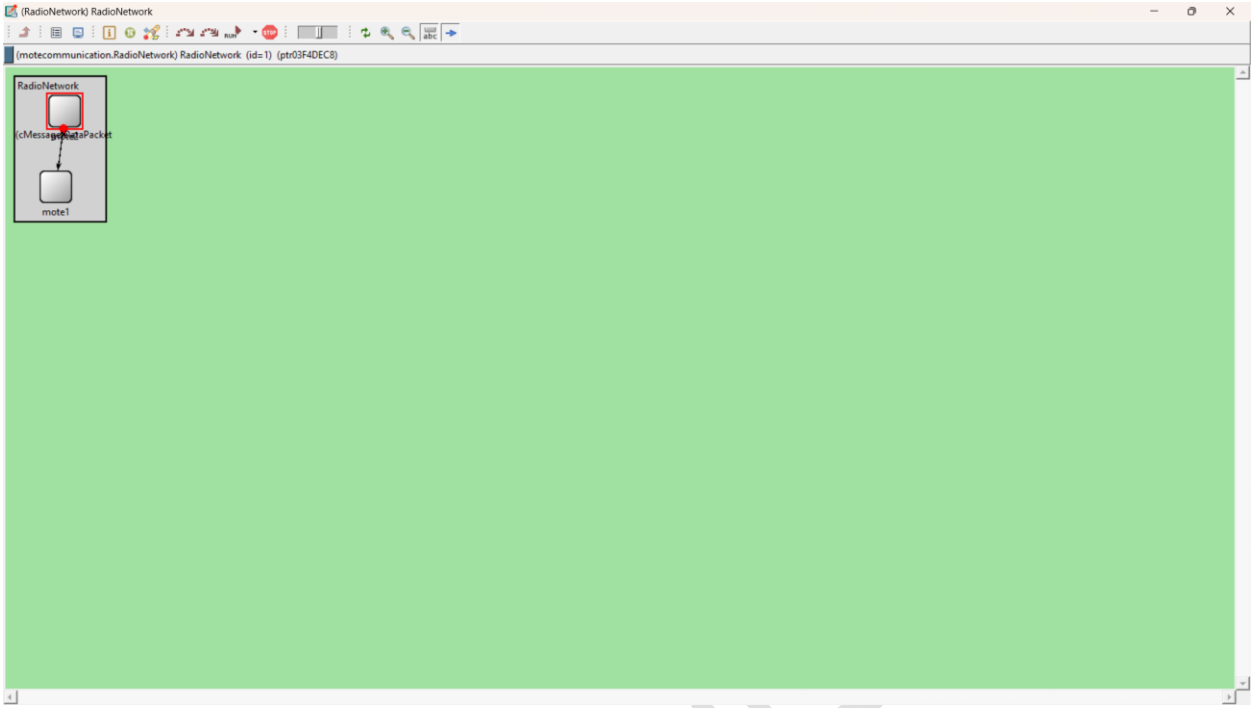
Network Topology (mote1 ↔ mote2)



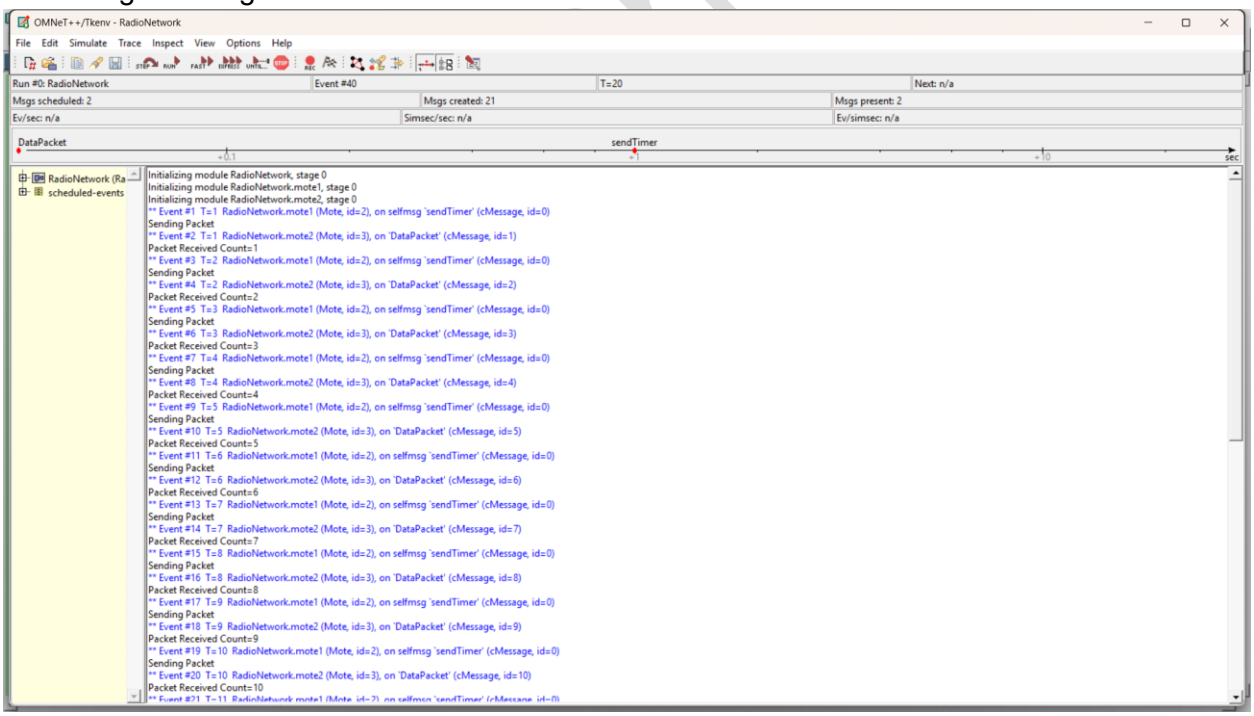
Jeetesh Shirganoor

Mvlu college
Wireless sensor network

Simulation Run:



Event Log Showing Packet Transmission:



Result

The packet transmission between two motes was successfully simulated. Packets sent by `mote1` were received by `mote2`, demonstrating mote-to-mote communication in the network.

Jeetesh Shirganoor T116